

Switch 6.0 Interchange Interface

Implementation Guide



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Preface

The "Switch 6.0 Interchange Interface Implementation Guide" describes the function and purpose of ACI's release 6.0 interfaces also called Switch 6.0 Interchange Interfaces and explains how customers can implement release 6.0 interfaces in their existing networks. Interchange interfaces that have been uplifted to a release 6.0 subvolume are referred to as Switch 6.0 throughout this implementation guide. Interchange interfaces that reside on a release 5.3 subvolume are referred to as Fortify.3 throughout this implementation guide. Interchange interfaces that reside on a release 5.1.2 subvolume are referred to as Fortify.1 throughout this implementation guide. This manual is intended to assist customers in configuring and controlling their Switch 6.0 Interchange Interfaces. Refer to the Switch 6.0 interface specific manual on the interface subvolume for details on how to configure and install the Switch 6.0 interface.

Software Release

The information in this manual applies to Switch 6.0 Interchange Interfaces running in BASE24 release 3.4/4.0, 5.0, 5.1.2, 5.3 and 6.0 networks.

Prerequisites

Users of this publication should be familiar with Interchange Interface processing. Knowledge of Tandem computer operations is also helpful.

Document Location

This document is located on the SW60UD04 subvolume in a file called NSTLSW60. It may be printed and copied as needed by ACI customers who have installed or intend to install ACI's Switch 6.0 Interfaces. The document may also be ordered, at a cost, through ACI Publications.

Document Control

Version Record

12/01, Version 1, Switch 6.0 Interchange Interfaces
01/02, Version 2, Switch 6.0 Interchange Interfaces
02/02, Version 3, Switch 6.0 Interchange Interfaces
03/02, Version 4, Fortify Interface Support in 6.0 MAKE Stream
04/02, Version 5, Support Switch 6.0 in Pre-Release 6.0 Network
06/02, Version 6, Transaction Security Services Support
07/02, Version 7, Support Switch 6.0 in Release 5.1.2 or Earlier
10/02, Version 8, Support Switch 6.0 in Release 5.1.2 or Earlier
01/03, Version 9, Discretionary Data Suppression
10/03, Version 10, Switch 6.0 Interchange Interfaces Version 4
11/04, Version 11, Switch 6.0 Interchange Interfaces Version 5
11/05, Version 12, Switch 6.0 Interchange Interfaces Version 6
11/06, Version 13, Switch 6.0 Interchange Interfaces Version 7

Document History

15JAN2002 tmh/173
Symptom: Section 2.1 IMPLEMENTATION OF SWITCH 6.0 IN RELEASE 5.1.2 OR EARLIER states "You must use the release 6.0 ICF requester and server."
Problem: The release 5.3 ICF requester and server must be used.
Fix: Modified text under ICF CONVERSION of section 2.1 to read 5.3 instead of 6.0.
Dependency: Restore NSTLSW60.
Reference: Cases 342371 and 342372

01FEB2002 tmh/173
Symptom: Unable to add/update data on screen 12 of the ICFE.
Problem: Segment 32 (CAF-ACCOUNT-SWI-DATA) in the SEG-TABLE of the PITABLE must be set to 'Y' in order to support switch specific data on screen 12 of the ICF/ICFE in release 6.0. This is documented in Section 4 of the Implementation Guide and should be documented in this document as well.
Fix: Added SEG-TABLE CHANGES within section 2.3 IMPLEMENTATION OF FORTIFY.1, FORTIFY.3 AND SWITCH 6.0 IN RELEASE 6.0.
Dependency: Restore NSTLSW60.
Reference: Case 343549

27FEB2002 tmh/173
Symptom: This document makes reference to BA60SC04.
Problem: This subvolume does not exist.
Fix: Modified the reference of BA60SC04 to BA10SC04.
Dependency: Restore NSTLSW60.
Reference: Case 344726

26FEB2002 TMH/173

Symptom: 1) Some Fortify interfaces dump at initialization when compiled in a release 6.0 MAKE stream.

 2) Some Fortify interface reports do not compile successfully when compiled in a release 6.0 MAKE stream.

Problem: 1) The interfaces dump when compiled in the release 6.0 MAKE stream. With the Format 2 File Support enhancement, procedure HISWFILE^OPEN in release 6.0 HISWUTLS was modified to invoke LIB_FILE_OPEN. LIB_FILE_OPEN is a kernel library call. Some Fortify interface MAKE files do not contain a dependency in the bind for OKLIB_OKLIBTO and do not contain a select search on OKLIB_OKLIBTO in the bind. Therefore, LIB_FILE_OPEN is not included in the executable objects of the interfaces.

 2) RP01SWIS and RP09SWIS were modified for the Format 2 File Support enhancement. As a result of these modifications, the 01 and 09 interface reports will no longer compile successfully when compiled in a release 6.0 MAKE stream.

Fix: 1) Added flag sw_make_fortify_60_on to the release 6.0 CUSTMACS. When set to TRUE, the MAKE files of the affected interfaces will include OKLIB_OKLIBTO in the dependency list of the bind and will include a select search of OKLIB_OKLIBTO in the bind. In any pre-release 6.0 MAKE stream, this flag will not exist and will default to the value of FALSE. When the affected interfaces are compiled in a pre-release 6.0 MAKE stream, the MAKE files will not include OKLIB_OKLIBTO in the dependency list of the bind and will not include a select search of OKLIB_OKLIBTO in the bind.

 2) Added flag sw_make_fortify_60_on to the release 6.0 CUSTMACS. When sw_make_fortify_60_on is TRUE, the MAKE files for the 01 and 09 interface reports will include OKLIB_OKLIBTO in the dependency lists. Added data to working storage that is required in order for the reports to compile successfully in release 6.0 MAKE stream.

 3) Added "Fortify Interface Support in release 6.0 MAKE Stream" to section Software Change Numbers (SCNS).

Dependency: Apply fix to CUSTMACS. Replace FLGSDOC and NSTLSW60.

Reference: WO #000202-01

16APR2002 tmh/173

Symptom: 1) Document states that the 5.3 ICF requester/server must be utilized when implementing Switch 6.0 in 5.1.2 or earlier.

 2) Release 5.0 Impacts and release 5.1.2 and 5.3 Impacts documentation is incorrect and incomplete.

 3) Implementing Switch 6.0 in 5.1.2 or earlier states that in order to support NCD, ensure that CRT Passthru and EFTPOS segments are set to 'N' in the PITABLE.

Problem: 1) A 5.1.2 customer will not have the 5.3 ICF requester/server.
2) The customer does not have to perform steps in order to enable them to extract a 6.0 token in a pre-release 6.0 network environment because there are currently no 6.0 tokens that will be logged in a pre-release 6.0 network environment.
3) There is no longer an EFTPOS segment in the 5.1.2 PITABLE.

Fix: 1) Modified verbiage to use the 6.0 ICF requester/server.
2) Added detailed installation instructions throughout the document and deleted installation steps that are unnecessary.
3) Modified verbiage to ensure that CRT Passthru segment is set to 'N' in order to support NCD in Switch 6.0 in 5.1.2 or earlier network.

Dependency: Restore NSTLSW60.
Reference: Case 346586

04JUN2002 tmh/173

Symptom: 1) Switch 6.0 Interfaces will not MAKE successfully when compiled with XPNET 2.1 in a pre-release 6.0 network environment.
2) Section Switch 6.0 Interface Implementation Considerations mentions two additional upgrades that the customer should consider.
3) Transaction Security Services Enhancement.

Problem: 1) Some XPNET globals were modified in XPNET 3.0 for release 6.0. Consequently, Switch 6.0 Interfaces no longer MAKE successfully with XPNET 2.1.
2) Only one additional upgrade is documented.
3) None.

Fix: 1) Added subsection Implementation of Switch 6.0 in XPNET 2.1 System that addresses this issue.
2) Added subsection XPNET Requirement to section Switch 6.0 Implementation Considerations.
3) Modified the document to include support of Transaction Security Services.
Modified sections: OVERVIEW
SWITCH 6.0 INTERFACES
Modified subsection: DDL IMPACTS
Added subsections: KEYF/KEY6 CONVERSION PROGRAMS
TRIPLE DES ENCRYPTION
REQUIREMENTS

Dependency: Restore NSTLSW60.
Reference: WO #020217-01

19JUL2002 tmh/173

Symptom: 1) In order to run Switch 6.0 Banknet, BIC ISO, MDS/ Cirrus ISO, MDS Maestro Gateway, PLUS ISO or VisaNet ISO Interface in a release 5.1.2 or earlier network environment, the 5.3 ICF conversion program must be run and the 5.3 ICF requester/server

must be utilized.

2) Section 2.1 IMPLEMENTATION OF SWITCH 6.0 IN RELEASE 5.1.2 OR EARLIER states "You must use the release 6.0 ICF requester and server.".

Problem: 1) A customer in a 5.1.2 or earlier network environment may not have release 5.3 code.

2) The release 5.3 ICF requester and server must be used.

Fix: 1) Removed all verbiage referencing ICF60M. This file was used along with B24C to automatically recompile all of the ICF requesters with the release 6.0 ICF DDL. Added verbiage to recompile all of the ICF requesters with the release 6.0 ICF DDL by recompiling all of the interface code in the release 6.0 MAKE stream. Also added verbiage in many places requesting that the customer request certain fixes if they are not current with fixes. The software order form has been updated so that 5.1.2 or earlier customers installing a Switch 6.0 Banknet, BIC ISO, MDS/Cirrus ISO, MDS Maestro Gateway, PLUS ISO or VisaNet ISO Interface may request the 5.3 code required to compile the 5.3 ICF requester/server and the 5.3 ICF conversion program. Created a new MAKE stream on BA60UC03 that will compile the 5.3 ICF conversion program and the 5.3 ICF requester/server.

2) Modified text under ICF CONVERSION of section 2.1 to read 5.3 instead of 6.0.

Dependency: Restore ICFMACC, ICFMACM, ICFMACS. Replace AAREAD03 and NSTLSW60.

Reference: Case #349852

16OCT2002 tmh/173

Symptom: 1) No documentation exists instructing how to support release 6.0 tokens in pre-release 6.0 network environments.

2) No documentation exists instructing how to install 6.0 LINK Interface in pre-release 6.0 network environments.

Problem: 1) No documentation exists instructing how to support release 6.0 tokens in pre-release 6.0 network environments.

2) No documentation exists instructing how to install 6.0 LINK Interface in pre-release 6.0 network environments.

Fix: 1) Added instructions on how to support release 6.0 tokens in a pre-release 6.0 network in section 2.2.

2) Added LINK to the list of interfaces in section 2.1.

Dependency: Restore NSTLSW60.

Reference: Case 354528

28JAN2003 gg/724
Symptom: Discretionary Data Suppression
Problem: None
Fix: Add documentation for Discretionary Data Suppression
 and Offline PIN Management. Update documentation to
 be current with new and uplifted interfaces for release
 6.0. Update interface lists for enhancements.
Dependency: Restore NSTLSW60.
Reference: WO #020625-02

21OCT2003 tmh/173
Symptom: Switch 6.0 Interfaces Version 4
Problem: None.
Fix: Modified documentation so that is current as of Release
 6.0 Version 4.
Dependency: Restore NSTLSW60.
Reference: WO #020625-02

08NOV2004 huac
Symptom: Switch 6.0 Interfaces Version 5
Problem: None.
Fix: Modified documentation so that is current as of Release
 6.0 Version 5.
Dependency: Restore NSTLSW60.
Reference: WO #040517-01

03NOV2005 hayest
Symptom: Switch 6.0 Interfaces Version 6
Problem: None.
Fix: Modified documentation so that is current as of Release
 6.0 Version 6.
Dependency: Restore NSTLSW60.
Reference: Case 402854

03NOV2006 hayest
Symptom: Switch 6.0 Interfaces Version 7
Problem: None.
Fix: Modified documentation so that is current as of Release
 6.0 Version 7.
Dependency: Restore NSTLSW60.
Reference: WO #419490

1: Introduction

Overview

A subset of ACI's Fortify.3 Interchange Interface modules were upgraded to Switch 6.0 Interchange Interfaces. Switch 6.0 Interchange Interfaces will process in a release 3.4/4.0, 5.0, 5.1.2, 5.3 or 6.0 BASE24 network.

ACI recommends that the BASE24 user be on a current Tandem-supported operating system to obtain maximum performance and support; however Switch 6.0 Interchange Interfaces are compatible with a D45 Compaq NSK operating system.

Following is a list of interchange interface subvolumes for Switch 6.0. The names of these subvolumes have been changed to SW60xxxx where "xxxx" identifies the interface. No new subvolumes were introduced for Switch 6.0 Version 7.

SW60AEGN	American Express Global Network
SW60AIF	Acquirer Information File
SW60AOI	Alaska Option ISO
SW60ATBI	BIC ISO ATM
SW60AXCI	American Express Credit ISO
SW60BAB1	BIC ISO Base
SW60BCGI	Boston Communications Group
SW60BIC	ANSI BASE24
SW60BNET	Banknet
SW60CKNY	NYCE ISO Check Extension
SW60CKVD	Visa DPS ISO Check Extension
SW60CSE	Switch CSE
SW60CSSI	Cash Station ISO
SW60DDSI	Deluxe ISO Generic
SW60DIAS	DIAS
SW60DIGI	Deluxe ISO Gateway
SW60DSCV	Discover ISO
SW60EBAO	Alaska Option ISO EBT Extension
SW60EBDG	Deluxe ISO Gateway EBT
SW60EBMP	Midwest Payment Systems Host to Host EBT Extension
SW60EURO	EPS-Net
SW60FDRI	First Data Resources ISO
SW60IAEG	American Express Global Network EMV Extension
SW60IBGC	NBGC EMV Extension
SW60IBNT	Banknet EMV Extension
SW60IEUR	EPS-Net EMV Extension
SW60ILNK	LINK EMV Extension
SW60IMDM	MDS Maestro Gateway EMV Extension

SW60IMDS	MDS/Cirrus ISO EMV Extension
SW60IPEP	IPF Refresh - EPS-Net
SW60IPEU	IPF Refresh - EPS-Net
SW60IPF1	IPF Refresh - Plus
SW60IPF2	IPF Refresh - Cirrus
SW60IPF3	IPF Refresh - VisaNet ISO
SW60IPST	IPF Refresh - Star ISO
SW60IPVD	IPF Refresh - Visa DPS ISO
SW60ISPN	SAMA SPAN2 EMV Extension
SW60ISWN	Switch Authorization Network EMV Extension
SW60IVIS	VisaNet ISO EMV Extension
SW60JCBI	JCB ISO
SW60LINK	LINK
SW60MACI	MAC MASM
SW60MAGI	Magic Line
SW60MDS	MDS/Cirrus ISO
SW60MDSM	MDS Maestro Gateway
SW60MEUR	EPS-NET U.K. Maestro Extension
SW60MONEY	Money Station
SW60MPS	Midwest Payment Systems Host to Host
SW60NBGC	NBGC
SW60NEN	NEN/Excel ISO
SW60NETI	Networks ISO
SW60NPCI	NPC ISO
SW60NPCM	NPC Multilink ISO
SW60NYCI	NYCE ISO
SW60PAYI	PayPoint
SW60PBLX	Publix
SW60PISI	Plus ISO Interlink Gateway
SW60PISO	Plus ISO
SW60PSBI	BIC ISO POS
SW60PULI	Pulse ISO
SW60RMT	Retail Message Translator
SW60RPT	Report
SW60SDF	Switch Dispute File
SW60SHZM	Shazam ISO
SW60SI92	Generic ISO Interchange
SW60SNF	Station Name File
SW60SPN2	SAMA SPAN2
SW60SRC	Switch Source
SW60STF	Switch Terminal File
SW60STRI	Star ISO
SW60STRX	Star ISO Disaster Recovery Extract
SW60SVS	SVS ISO
SW60SWAN	Switch Authorization Network
SW60SWCH	Generic Switch

SW60SWPM	Switch Performance Monitoring
SW60VDPS	Visa DPS ISO
SW60VISA	VisaNet ISO
SW60VISI	VisaNet Interlink
SW60VISP	VisaNet Plus Gateway
SW60VLNK	ValueLink
SW60XML	Generic XML Switch
SW60XTI	XML Telco

Assumptions

This manual assumes that ACI customers who migrate to Switch 6.0 Interfaces will use the Fortify.3 Interfaces as a base starting point. File conversions required for migration to the Fortify.3 Interfaces will not be addressed. Please refer to the Fortify.3 implementation guide for further information.

Other Documents

In addition to this guide, the reader will want to obtain other BASE24 publications which cover operational and product specific information required to fully understand and implement the release 6.0 software. Such manuals should initially include:

- BASE24 Classic 6.0 Installation Guide - BA-AE000-54

Describes how to install a release 6.0 system.

- Switch specific manuals and update files on switch subvolumes

Contain information specific to the interchange interface.

- BASE24 Release 6.0 Implementation Guide - BA-AE000-81

Describes the differences between release 5.3 and release 6.0.

- ACI MAKE Release 5 User Guide - TO-MK000-06

Introduces ACI's MAKE tool. The documentation includes an overview on MAKE and Macros used with ACI's MAKE tool.

- BASE24 Tokens Manual - BA-AE000-12

A reference guide to the token-based internal message used by BASE24.

- BASE24 Base Files Maintenance Manual - BA-AE000-03

Documents the file maintenance screens used to access the files and functions shared among all BASE24 products.

- NET24-XPNET Release 3.0 Documentation - CD-AE000-07

Describes how to install an XPNET 3.0 system.

Switch 6.0 Interfaces

Switch 6.0 Interfaces, like the Fortify.3 and Fortify.1 Interfaces, will allow ACI to maintain, support and update only one interface module per interchange as opposed to one module for release 3.4/4.0, 5.0, 5.1.2, 5.3 and 6.0. This allows ACI to be more responsive to switch mandates. In addition, customers using Switch 6.0 Interfaces only need to support one interface module per interchange which will run in all releases of BASE24. Since these interfaces will run in multiple releases, it will allow customers to utilize a phased migration strategy while moving to the most current release of BASE24.

Switch Mandates and Fixes

Implementing Switch 6.0 Interfaces allows customers to stay current with changes made to interchange interfaces. Switch 6.0 Interfaces are made current with fixes with each mandate delivery.

Some Switch 6.0 Interfaces were modified to support the following enhancements:

- Multinational Transaction Enhancement
- Check Authorization Enhancement
- Transactions Allowed Enhancement
- EMV Enhancement
- Triple/Single DES Enhancement
- Dynamic Key Management Support via Transaction Security Services
- Triple DES Encryption (3DEA) Support via Transaction Security Services
- Discretionary Data Suppression Enhancement
- Offline PIN Management Enhancement
- Format 2 Files Enhancement
- Triple-Length Key Support via Transaction Security Services
- Forward Declined Advices
- Contactless Chip and Magnetic Stripe
- Acquirer Information File (AIF)

The Format 2 Files, EMV, Transactions Allowed Transaction Security Services, Triple-Length Key Support via Transaction Security Services and Offline PIN Management Enhancements are only supported in a BASE24 release 6.0 environment. The Format 2 Files Enhancement requires a D46 or higher Compaq NSK operating system. The Triple/Single DES Enhancement is supported in any BASE24 pre-release 6.0 network environment. Discretionary Data Suppression is supported in Interchange Interfaces in a BASE24 release 5.3 or 6.0 environment, and in POS Router Auth in release 6.0 only.

Refer to the interface manual or the interface update file on the respective interface subvolume in order to determine what enhancements the interface supports. The

manual may be found on the subvolume with the name xxxxMAN or xxxxMNWD where "xxxx" identifies the interface. The update files are named xxxxUPDT where "xxxx" identifies the interface.

For example, the documentation for enhancements to the VisaNet ISO Interface is contained in a file named VISAUPDT and the documentation for enhancements to the Banknet Interface is in a file named BNETUPDT.

Phased Migration to Switch 6.0 Interfaces

Customers have the option of migrating from Fortify.3 Interfaces to Switch 6.0 Interfaces all at once or one at a time. File conversion programs are provided to allow customers to make Fortify.3 Interfaces compatible with Switch 6.0 Interfaces.

Switch 6.0 requires a minimum of a D45 Compaq NSK operating system. Customers may individually phase each Switch 6.0 Interface into their existing environment.

Recertification Requirements

Recertification may be required for the Switch 6.0 Interfaces. The decision to recertify with the interchange is ultimately the decision of the customer and the interchange.

Switch 6.0 Interface Implementation Consideration

There are upgrades that customers should consider while implementing their Switch 6.0 Interfaces in order to make their eventual upgrade to release 6.0 easier.

- Tandem Operating System Requirements
- XPNET Requirements
- Triple DES Encryption (3DEA) Requirements

Tandem Operating System Requirements

ACI recommends that the BASE24 user be on a current Tandem-supported operating system to obtain maximum performance and support; however Switch 6.0 Interchange Interfaces are compatible with a D45 Compaq NSK operating system. The Format 2 Files Enhancement requires a D46 or higher Compaq NSK operating system.

XPNET Requirements

Release 6.0 requires an XPNET 3.0 system.

Triple DES Encryption (3DEA) Requirements

Support of triple DES encryption via Transaction Security Services requires the Racal HSMs to be an RG7000 with revision 1.05/5.05 firmware. The enhancement also requires the Atalla HSMs to be an A9000 or A10000 with revision 2.9 or Atalla AKB 1.0 firmware.

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2: Implementation and Configuration

Introduction

This section covers the following topics concerning the implementation and configuration of Switch 6.0 Interfaces:

- Restore
- Software Change Numbers (SCNS)
- MAKE
- Implementation of Switch 6.0 in release 5.3, 5.2.1 or earlier.

If you are running a Switch 6.0 Interface in a release 5.1.2 or earlier network environment, refer to subsections 2.1 and 2.2.

If you are running a Switch 6.0 Interface in a release 5.3 network environment, refer to subsections 2.2.

- Implementation of Fortify.1, Fortify.3 or Switch 6.0 in release 6.0.

If you are running a Fortify.1 Interface in release 6.0, refer to subsection 2.3.

If you are running a Fortify.3 Interface in release 6.0, refer to subsection 2.3.

If you are running a Switch 6.0 Interface in release 6.0, refer to subsections 2.3 and 2.4.

- System impacts
- File conversion programs
- File impacts

NOTE: Installation steps are depicted in bold text.

Restore

Restore the software tape to the appropriate volume and subvolumes. In order to restore the tape, enter the following command at the TACL prompt:

```
>RESTORE /OUT $S.#REST, NOWAIT/ $TAPE,(*. *.*), VOL $<vol>,  
LISTALL, OPEN, MYID, TAPEDATE
```

where \$<vol> is the volume where you want the files restored.

Repeat this step for each software tape.

Detailed instructions on how to restore the Switch 6.0 software tapes can be found in the BASE24 Classic 6.0 Installation Guide document (BA-AE000-54). The Switch 6.0 tapes are a subset of the BASE24 release 6.0 code.

Note: Only those sections that pertain to Base and Switch Interfaces need to be referenced in the Install document. An example would be that the sections RESTORE SOFTWARE TAPES and COMPILATION STEPS may need to be referenced. However, the section APPLY FIXES would not need to be referenced.

Software Change Numbers (SCNS)

Implementing Fortify.1 Interfaces in Release 6.0

In order to successfully make some of the Fortify.1 Interfaces in a release 6.0 network environment, utilizing the release 6.0 MAKE stream, the following fixes must be applied to the following Fortify.1 Interface files. If you are not current with fixes, please request the fix(es) you are missing.

<u>FILE</u>	<u>SCN#</u>
SW51AOS8.SVAMLFS	SW5F788
SW51AS82.SVMLFS	SW5F789
SW51DDSI.SVDMLFS	SW5F790
SW51DDSI.SVDPMLFS	SW5F791
SW51MAGI.SVMMLFS	SW5F792
SW51SHZM.SVSMLFS	SW5F793

Implementing Switch 6.0 Interfaces in Release 5.3

In order to successfully make a Switch 6.0 Interface in a release 5.3 network environment, utilizing the release 6.0 MAKE stream, the following fixes must be applied to the following files. If you are not current with fixes, please request the fix(es) you are missing.

<u>FILE</u>	<u>SCN#</u>
BA53DDL.DDLFCNF	BA5P392
BA60AFT.BAAFTM	BA05098
BA60EXT.EXTRM	BA05099

Implementing Switch 6.0 Interfaces in Release 5.1.2

In order to successfully make a Switch 6.0 Interface in a release 5.1.2 network environment, utilizing the release 6.0 MAKE stream, the following fixes must be applied to the following files. If you are not current with fixes, please request the fix(es) you are missing.

<u>FILE</u>	<u>SCN#</u>
BA51DDL.DDLFCNF	BA5B687
BA60AFT.BAAFTM	BA05098
BA60EXT.EXTRM	BA05099

Implementing Switch 6.0 Interfaces in XPNET 2.1 System

In order to successfully MAKE a Switch 6.0 Interface in a pre-release 6.0 network environment with XPNET 2.1, a new file, GLOBAL30, located on \$vol.SPANNET, is required.

Fortify Interface Support in Release 6.0 MAKE Stream

In order for the Fortify.1 and Fortify.3 Interfaces to execute successfully when compiled in a release 6.0 MAKE stream, the following fixes must be applied. If you are not current with fixes, please request the fix(es) you are missing.

<u>FILE</u>	<u>SCN#</u>
BA60MAKE.CUSTOMACS	BA05301
BA60MAKE.FLGSDOC	BA05302
SW51MACI.MACIM	SW5F877
SW51MONY.MONYM	SW5F886
SW51NEN.NENM	SW5F880
SW51NEN.NENLIBM	SW5F881
SW51NET8.NET8M	SW5F891
SW51PADP.ADPM	SW5F888
SW51PBLX.PBLXM	SW5F887
SW53AMXI.AMXIM	SW5P703
SW53BIC.BICM	SW5P698
SW53CIRR.SWCIRRM	SW5P708

In order for the Fortify.1 and Fortify.3 Interface 01 and 09 reports to compile successfully when compiled in a release 6.0 MAKE stream, the following fixes must be applied. If you are not current with fixes, please request the fix(es) you are missing.

<u>FILE</u>	<u>SCN#</u>
BA60MAKE.CUSTOMACS	BA05301
BA60MAKE.FLGSDOC	BA05302
SW51MACI.RP09MACM	SW5F878
SW51MACI.RP09MACS	SW5F879
SW51NEN.RP01NENM	SW5F882
SW51NEN.RP01NENS	SW5F883
SW51NEN.RP09NENM	SW5F884
SW51NEN.RP09NENS	SW5F885
SW51NET8.RP01NETM	SW5F892
SW51NET8.RP01NETS	SW5F893
SW51NET8.RP09NETM	SW5F894
SW51NET8.RP09NETS	SW5F895

SW51PADP.RP09ADPM	SW5F889
SW51PADP.RP09ADPS	SW5F890
SW53AMXI.RP01AMIM	SW5P704
SW53AMXI.RP01AMIS	SW5P705
SW53AMXI.RP09AMIM	SW5P706
SW53AMXI.RP09AMIS	SW5P707
SW53AXCR.RP09AXCM	SW5P713
SW53AXCR.RP09AXCS	SW5P714
SW53BIC.RP01BICM	SW5P699
SW53BIC.RP01BICS	SW5P700
SW53BIC.RP09BICM	SW5P701
SW53BIC.RP09BICS	SW5P702
SW53CIRR.RP01CIRM	SW5P709
SW53CIRR.RP01CIRS	SW5P710
SW53CIRR.RP09CIRM	SW5P711
SW53CIRR.RP09CIRS	SW5P712

Detailed instructions on how to apply fixes can be found in the BASE24 Classic 6.0 Installation Guide document (BA-AE000-54).

MAKE

MAKE is a utility that enables automated compilation of BASE24 modules and enforces module dependencies. It was introduced with release 5.1 and the Fortify.1 interfaces. It replaces traditional BASE24 compile files and procedures. Previously, the compile commands for each module had to be initiated manually, and the completion status of each compilation had to be checked manually. The MAKE utility automatically compiles each module, if necessary, and checks the compilation completion status. The MAKE utility also enforces module dependencies and automatically compiles modules if any of these dependencies have changed. For example, when a DDL is changed, the MAKE utility will ensure all the modules that use the DDL are compiled as well. This greatly enhances the customer's ability to manage BASE24 software. For more information on MAKE, see "ACI MAKE Release 5 User Guide", TO-MK000-06.

BASE24 will be compiled via MAKE. For more information on how to compile, see COMPILATION STEPS in the "BASE24 Classic 6.0 Installation Guide", BA-AE000-54.

2.1 Implementation of Switch 6.0 in Release 5.1.2 or Earlier

IMPORTANT

The following installation steps are required only if implementing Switch 6.0 Banknet, BIC ISO, LINK, MDS/Cirrus ISO, MDS Maestro Gateway, PLUS ISO or VisaNet ISO Interfaces in a 5.1.2 or earlier BASE24 network environment and are not currently using the 5.3 ICF and 5.3 ICF requester/server.

ICF Requirements

Installing Switch 6.0 Banknet, BIC ISO, LINK, MDS/Cirrus ISO, MDS Maestro Gateway, PLUS ISO or VisaNet ISO Interface in release 5.1.2 or earlier requires the utilization of the 5.3 ICF requester/server, conversion of the current ICF to the release 5.3 ICF and recompilation of the current interface requesters with the release 6.0 ICF DDL.

ACI provides the following subvolumes that are necessary to compile and run the standard 5.3 ICF conversion program and to compile the 5.3 ICF requester/server:

- BA10SKEL
- BA53AFT
- BA53CSE
- BA53DDL
- BA53DEFS
- BA53OBJ
- BA53PTHE
- BA53SRC
- BA53UC03
- BA60UC03

Customers are not sent all of the BASE24 Release 5.3 code, so the standard BASE24 Release 5.3 MAKE stream cannot be used. Therefore, in addition to the necessary release 5.3 subvolumes, a special Make subvolume, BA60UC03, is provided that is used to compile and build the standard 5.3 ICF conversion program and to compile the 5.3 ICF requester/server. For more information about the files on the special Make subvolume, BA60UC03, refer to the AAREAD03 file located on the subvolume.

A MAKE stream resides on BA60UC03 that creates the release 5.3 ICF requester/server and the 5.3 ICF conversion program. Do not run EQUIV on

this MAKE stream. Perform the following steps using the editor of your choice (e.g., EDIT):

> VOLUME BA60UC03

Replace instances of the system name \global with the appropriate system name in ICFMACC, ICFMACM and ICFMACS.

Replace instances of the volume name \$bar60 with the appropriate volume name in ICFMACC, ICFMACM and ICFMACS.

If you do not have the volume \$SYSACI, replace instances of the volume name \$SYSACI with the appropriate volume in ICFMACC, ICFMACM and ICFMACS.

> O ICFMACC

Scup the release 5.3 ICF requester object into the current POBJ file.
Point the SERVER-ICF to the release 5.3 ICF server object file.

In order to remake all interface requesters using the 6.0 ICF DDL, simply recompile all of your interfaces in the release 6.0 MAKE stream. Scup all of the new interface requester objects into the current POBJ file.

Create a new ICF data file. (NOTE: The SET EXT will depend on the number of ICF records you have. Refer to your existing ICF data file for the value to use for your site.)

> VOLUME <lgnt>DATA

Where <lgnt> is the name of the logical net, i.e., tes1, pro1, cer1.

> FUP RENAME ICF*,OLDDATA.*

> FUP

RESET

```
SET ALTKEY ("AK", KEYOFF 10, KEYLEN 16, FILE 0, UNIQUE)
SET NO ALTCREATE
SET ALTFIL ( 0, ICF0 )
SET TYPE K
SET KEYOFF 6
SET KEYLEN 20
SET REC 1660
SET BLOCK 4096
SET IBLOCK 4096
SET EXT( 4, 32 )
SET MAXEXTENTS 100
```

```
CREATE ICF
  RESET
  SET TYPE K
  SET KEYLEN 18
  SET REC 38
  SET BLOCK 4096
  SET IBLOCK 4096
  SET EXT( 4, 32 )
  SET MAXEXTENTS 100
CREATE ICF0
```

Convert the current ICF to the release 5.3 ICF. Refer to the run file BA53UC03.ICFR for the ICF conversion program execution details.

Non-Currency Dispense (NCD)

In order to support NCD in the Switch 6.0 Interfaces (MDS/Cirrus ISO, MDS Maestro Gateway, PLUS ISO and VisaNet ISO), the NCD segment must be turned off.

> EDIT BA51AFT.PITABLE

In order to ensure that the NCD segment is turned off, verify that the current PITABLE has the CRT-Passthru segment set to "N".

NOTE: If it is necessary to modify your current PITABLE, you will need to remake RQMEGA and SCUP the newly made RQMEGA object (RQMEGAO) into your current POBJ file. You must continue to use your current RQMEGA and PITABLE files. Do not use the release 6.0 RQMEGA or PITABLE.

For the above listed Switch 6.0 Interfaces, NCD issuer transactions will be processed as ATM withdrawals when the NCD segment does not exist. Customers may want to adjust the limits for ATM withdrawals to reflect their NCD traffic. NCD acquirer transactions will come in as withdrawals. If the NCD token (A5) is present, the transaction processing code will be modified to reflect a purchase.

2.2 Implementation of Switch 6.0 in Release 5.3, 5.1.2 or Earlier

IMPORTANT

The following information is necessary when running a Switch 6.0 Interface or a combination of Switch 6.0 and Fortify.1/Fortify.3 Interfaces in a pre-release 6.0 BASE24 network environment.

MAKE

For pre-release 6.0 BASE24 network environments supporting a Switch 6.0 Interface or a combination of Switch 6.0 and Fortify.1/Fortify.3 Interfaces, it is a requirement that the Switch 6.0 Interface be made with the pre-release 6.0 LCONF DDL. In order to do so, you must edit BA60MAKE.CUSTOMACS and set flag SW_PRE_60_LCONF_ON to TRUE. You must edit BA60DDL.BADDLMM and ensure that the baddl_ddlconf define points to the location of the DDLFCNF file in your BASE24 network environment (i.e., BA53DDL.DDLFCNF, BA51DDL.DDLFCNF or BA50DDL.DDLFCNF).

For pre-release 6.0 BASE24 network environments supporting Switch 6.0 Banknet, EPS-Net, MDS/Cirrus ISO, MDS Maestro Gateway or VisaNet ISO Interfaces, the respective CUSTOMACS flags BNET_EMV_ON, EURO_EMV_ON, MDS_ISO_EMV_ON, MDS_MAESTRO_EMV_ON and VISA_EMV_ON must be set to FALSE. EMV processing is only supported in the above mentioned Switch 6.0 Interfaces in a release 6.0 BASE24 network environment.

For pre-release 6.0 BASE24 network environments supporting Switch 6.0 BIC ISO, Banknet, MDS/Cirrus ISO, MDS Maestro, PLUS ISO or VisaNet ISO Interfaces, the respective CUSTOMACS flags BIC_ISO_ENHANCED_MODE_ON, BNET_ENHANCED_MODE_ON, MDS_ENHANCED_MODE_ON, PLUS_ISO_ENHANCED_MODE_ON and VISA_ENHANCED_MODE_ON must be set to FALSE.

You must make your Switch 6.0 Interface code in the release 6.0 MAKE stream. Whenever you need to re-make existing switch interface code or whenever you install a new Fortify.1, Fortify.3 or Switch 6.0 Interface, the switch interface files MUST be made in the release 6.0 MAKE stream.

> EDIT BA60MAKE.CUSTOMACS

1. Set SWITCH_ON to true.
2. Set all interface flags to true that are currently set to true in your current BA5*MAKE.CUSTOMACS. Set flags to true for any new interfaces that

you have purchased. Do not set any of the EMV or ENHANCED related flags to true.

3. Do not enable the ATM, POS, etc. flags. The CLASSIC_ON and BASE_ON flags should already be set to true.
4. Change node and volume references to reflect your site.
5. Set appropriate compiler options.
6. Set object file destination flags.
7. Read the installation section in each of the XXXXMAN files on the interface subvolumes to determine which flags need to be set to true.

> EDIT BA5*MAKE.CUSTMACS

Set all switch flags to false since all interfaces will be compiled via the release 6.0 MAKE stream.

> EDIT BA60DDL.BADDLM

Change the baddl_ddlconf define to point to the location of your ddlconf, (ba51ddl.ddlconf or ba53ddl.ddlconf).

Make sure that the following files are on your system in order to compile BASE24 using ACI's MAKE:

```
$<vol>.BA60MAKE.ALLMACS  
$<vol>.BA60MAKE.CUSTMACS  
$<vol>.BA60MAKE.EQUIV  
$<vol>.MAKEV500.MAKEV501  
$<vol>.MAKEV500.MAKEHELP  
$<vol>.MAKEV500.TACLINIT  
$<vol>.MAKEV500.TACLMACS
```

MAKE will use BA60MAKE.EQUIV to change the ACI MAKE files to point to the correct system, volume, and subvolume.

The left half of the EQUIV file lists the system, volume, and subvolume of the system the BASE24 files came from.

> EDIT BA60MAKE.EQUIV

Change the right side of the EQUIV file to reflect the file locations on your system.

MAKEV500.MAKEV501 has some new features. In BA60MAKE.B24C, there are six examples. The first example updates all of your BASE24 MAKE files to point to the correct system, volume, and subvolumes. (This replaces the MAKESCAN that was used on earlier versions of MAKE.

> VOLUME BA60MAKE

> EDIT B24C

Set the correct volume locations.

Uncomment the following statement:

```
RUN $<vol>.MAKEV500.MAKEV501/OUT $ S.#MAKE.B24M/  
$<vol>.BA60MAKE.CUSTMACS -DETAIL -UPDATE
```

> OBEY B24C

Field Support recommends running MAKE always using the DETAIL option. If the SUMMARY option is used instead, you cannot determine which files were not found. NOTE: If MAKE abends, correct the problem and restart MAKE.

Additionally, you may receive a warning from MAKE that the MAX_RADIX_TBL_SIZE is too big and a recommendation on what the size should be. Use this value in your B24C file. If the specified size is too small, MAKE will abend.

After MAKE has completed successfully, re-add the comments in front of the first run statement.

For the purpose of an initial install, compile the entire system using BA60MAKE.B24C. The second run statement will check whether all of the MAKE dependencies are in place.

> VOLUME BA60MAKE

> EDIT B24C

Uncomment the following statement:

```
RUN $<vol>.MAKEV500.MAKEV501/OUT $S.#MAKE.B24M/  
$<vol>.BA60MAKE.B24M -FORCE -NOBUILD
```

> OBEY B24C

After MAKE has completed successfully, re-add the comments in front of the second run statement.

Optional MAKE Run

To verify that there aren't any targets, uncomment the third run statement.

> EDIT B24C

Uncomment the following statement:

```
RUN $<vol>.MAKEV500.MAKEV501/OUT $S.#MAKE.B24M/  
$<vol>.BA60MAKE.B24M -FORCE -PURGE_TARGETS
```

> OBEY B24C

After MAKE has completed successfully, re-add the comments in front of the third run statement.

The fourth run statement uses the `TIMESTAMP` and `NO_STOP_ON_ERRORS` flags. REMOVE the `NO_STOP_ON_ERRORS` statement and add the `SUPPRESS_UNTIL_SERVER` statement.

> VOLUME BA60MAKE

> CREATE TSTEMP

The modification timestamp on this file will be used by MAKE to determine if a file needs to be rebuilt. Edit `BA60MAKE.B24C` to verify it is set up like the example below:

> EDIT B24C

Uncomment the following statement and modify as follows:

```
RUN $<vol>.MAKEV500.MAKEV501/OUT $S.#MAKE.B24M/  
$<vol>.BA60MAKE.B24M -LIST -LISTREPLACE -  
TIMESTAMP_FILE=$<vol>.BA60MAKE.TSTEMP -  
SUPPRESS_UNTIL_SERVER
```

> OBEY B24C

Modify the switch interface processes within your network to point to the new object files.

LCONF Parameters

BASE24-REL

You must ensure that LCONF parameter BASE24-REL exists and is set to 40 for BASE24-atm release 4.0 and/or BASE24-pos release 3.4, 50 for BASE24 release 5.0/5.1.2/5.3 or 60 for BASE24 release 6.0. If this LCONF parameter does not exist, the Switch 6.0 Interface will default this parameter to 60 and will not process correctly.

DES-TRIPLE-SINGLE

If you are running a Switch 6.0 Interface that has been modified to support the Triple/Single DES Enhancement, (Banknet, BIC ISO, MDS/Cirrus ISO, MDS Maestro Gateway or VisaNet ISO), and you want the interface to support triple DES encryption, the LCONF parameter DES-TRIPLE-SINGLE must exist and be set to 'Y'. Single DES encryption processing will be performed if the parameter is missing or invalid.

If you are running a Switch 6.0 Interface that has been modified to support the Triple/Single DES Enhancement and do not want the interface to support triple DES encryption, LCONF parameter DES-TRIPLE-SINGLE is not required or may be added and set to 'N'.

LCONF Assigns

KEY6

If you are running a Switch 6.0 Interface that has been modified to support the Triple/Single DES Enhancement, (Banknet, BIC ISO, MDS/Cirrus ISO, MDS Maestro Gateway or VisaNet ISO), and you want the interface to support triple DES encryption, the KEY6 assign must be present in the LCONF. This assign is required when the DES-TRIPLE-SINGLE LCONF parameter is present and indicates triple DES encryption is to be performed. The interface will fail to initialize if the KEY6 assign is missing when the parameter DES-TRIPLE-SINGLE indicates triple DES encryption.

KEYF

If you are running a Switch 6.0 Interface that has been modified to support the Triple/Single DES Enhancement, (Banknet, BIC ISO, MDS/Cirrus ISO, MDS Maestro Gateway or VisaNet ISO), and you do not want the interface to support triple DES encryption, the KEYF assign must be present in the LCONF. This assign is required when single DES encryption is to be performed. The interface will fail to initialize if the KEYF assign is missing when the parameter DES-TRIPLE-SINGLE indicates single DES encryption or when parameter DES-TRIPLE-SINGLE is missing.

Security Table (SECTBL) Changes

If you are running a Switch 6.0 Interface that has been modified to support the Triple/Single DES Enhancement, (Banknet, BIC ISO, MDS/Cirrus ISO, MDS Maestro Gateway or VisaNet ISO), and you want the interface to support triple DES encryption, the KEY6 file will be used. In release 6.0, entries have been added to the BA60AFT.SECTBL in order to support the screens for the KEY6 requester/server. You must manually make the release 6.0 SECTBL modifications to your SECTBL and must re-make your RQMEGAS and SVISECS.

If you are running a Switch 6.0 Interface that has been modified to support the Triple/Single DES Enhancement and do not want the interface to support triple DES encryption, no SECTBL modifications are required.

6.0 Token In a 5.0/5.1.2/5.3 Environment

To enable selecting the new tokens for extracting or host transmission, customers should SCUP COPY the release 6.0 token requester (TKN) to the network's POBJ file and bind the release 6.0 token conversion utility into Extract and/or HISO.

The release 5.0 Extract and HISO programs should be rebound using the release 6.0 token conversion program. In both BA50HISO.HISOB and BA50EXT.EXTRB the line "select search BA50SRC.TKNCV" should be changed to "select search BA60SRC.TKNCV".

The release 5.1.2/5.3 Extract and or HISO programs should be rebound using the release 6.0 token conversion program. This may be accomplished by running MAKE after overriding the "basrc_tkncv" define in 5.1.2/5.3 CUSTMACS to point to the release 6.0 BA60SRC subvolume. To override this define the basrc_tkncv definition must be inserted to the CUSTMACS file.

Note: There are many different ways to override a define, but ACI recommends that these special overrides be kept in the 5.1.2/5.3 CUSTMACS file.

Other workaround alternatives are:

- 1) Update the release 5.0 DDLBATKN, DDLATTKN and DDLPSTKN files with the release 6.0 token definitions.
- 2) Update the release 5.0 BATKNID, ATTKNID and PSTKNID files with the release 6.0 token IDs.
- 3) Update the release 5.0 COBNAMES file by:
 - a) Adding the release 6.0 token IDs in the TKN-TABLE in section TKN-TABLE.
 - b) Increment the WS-TKN-TBL and TKN-TABLE-CNT for each token added.

- 4) Recompile the token requester (RQTKNS on the BA50AFT subvol).

- * Currently, the Interface will log all tokens if a token record does not exist. Also, if a token record does exist, the interface will log all tokens whose token IDs are not in the token record. If sending specific release 6.0 tokens to the host/co-network or logging these tokens to the ILF and/or PTLF/TLF is not desired, perform the following:

Add token records with the TRAN LOG fields corresponding to the specific release 6.0 token IDs set to "N". By adding a token record, only the tokens whose token IDs are included in the token record with their corresponding TRAN LOG fields set to "Y" will be logged to the ILF and PTLF/TLF. Since these token IDs within the token record have TRAN LOGs of "N", they will not be included in either log file.

- * If sending release 6.0 tokens to the host/co-network or logging these tokens to the ILF and/or PTLF/TLF is desired, perform the following:

Update the token conversion utilities to include the conversions written for the release 6.0 ATM tokens (ATTKNCVS on the BA50SRC subvolume), the release 6.0 base tokens (BATKNCVS on the BA50SRC subvolume) and the release 6.0 POS tokens (PSTKNCVS on the BA50SRC subvolume). Recompile the BASE token conversion utilities, then rebind HISO and EXTRACT. By updating the conversion utilities, extract will be able to process the release 6.0 tokens logged to the ILF and/or PTLF/TLF.

Once these issues are resolved, the token record can be updated to include/exclude the release 6.0 tokens for the host and/or the interchange and the extract process will be able to process the tokens should they be logged to the ILF and/or PTLF/TLF.

2.3 Implementation of Fortify.1, Fortify.3 or Switch 6.0 in Release 6.0

IMPORTANT

The following information is necessary when running a Fortify.1, Fortify.3 or Switch 6.0 Interface in a release 6.0 BASE24 network environment.

SEG-TABLE Changes

In a release 6.0 BASE24 network environment, segment 32 (CAF-ACCOUNT-SWI-DATA) must be set to 'Y' in the PITABLE in order to support interchange specific data on screen 12 of the ICF and ICFE.

> EDIT BA60AFT.PITABLE

MAKE

In a release 6.0 BASE24 network environment, all code must be compiled with the release 6.0 LCONF DDL. Therefore, CUSTMACS flag SW_PRE_60_LCONF_ON must be set to FALSE.

All interface code (Fortify.1, Fortify.3 and Switch 6.0) must be compiled in the release 6.0 MAKE stream. The interface objects created by the release 6.0 MAKE must be utilized in the release 6.0 network environment.

LCONF Parameter

BASE24-REL

You must ensure that LCONF parameter BASE24-REL exists and is set to 60. If this LCONF parameter does not exist, the Fortify.1 and Fortify.3 Interfaces will default this parameter to 40 and will not process correctly.

ENTIRE SYSTEM

If the switch install is part of an entire system install, see "BASE24 Classic 6.0 Installation Guide", BA-AE000-54, "BASE24 Release 6.0 Implementation Guide", BA-AE000-81 and the Word document, "BASE24 Release 6.0 Implementation Guide".

NEW AND CHANGED MODULES

To install switches in an existing BASE24 system, edit the CUSTMACS file or use MAKESCAN to change the location of the files, objects and modules.

To compile all changed and new modules volume to \$vol.BA60MAKE and set the appropriate flags in CUSTMACS. Volume to \$vol.BA60SRC and run the following command.

```
MAKEVxxx/OUT $S.#MAKE, NOWAIT, NAME $MAKE/B24M -NOLIST
```

where: xxx is the current MAKE version

To force compile an entire BASE24 system run the following command.

```
MAKEVxxx/OUT $S.#MAKE, NOWAIT, NAME $MAKE/B24M -NOLIST  
-FORCE
```

where: xxx is the current MAKE version

CONVERSION FILES

If customers find it necessary to compile a conversion file program, they will need to edit the specific MAKE file to change the volume and subvolume references to reflect the file locations on their system.

INDIVIDUAL MODULES

To compile a particular switch interface, follow the instructions in the manual for that switch. The manual can be found on the interface subvolume with the name xxxxMAN where "xxxx" identifies the interface. ACI recommends always compiling using the B24M MAKE file to ensure all dependent modules are compiled and bound.

XPNET 3.0

In an XPNET 3.0 system, files that were in the SPANNET and SPAN subvolumes for an XPNET 2.1 or earlier system have been moved to the XPNET subvolume. The define names for these files will not change but must be modified to point to the XPNET subvolume in the \$(VOL).BA60SRC.ALLMACS file. For example, SPANNET_SKEL will point to \$(VOL).XPNET.SKEL.

If the EQUIV file is not created by MAKESCAN using the EQUIV option but is used by MAKESCAN with the UPDATE option, the \$SYSACI.SPAN line should be removed and SPANNET must be changed to XPNET before running MAKESCAN. If MAKESCAN creates the EQUIV file using the EQUIV option, no change is necessary.

2.4 Implementation of Switch 6.0 in Release 6.0

IMPORTANT

The following information is necessary when running a Switch 6.0 Interface in a release 6.0 BASE24 network environment.

MAKE

For a release 6.0 BASE24 network environment supporting Switch 6.0 BIC ISO, Banknet, MDS/Cirrus ISO, MDS Maestro, PLUS ISO or VisaNet ISO Interfaces, the respective CUSTMACS flags BIC_ISO_ENHANCED_MODE_ON, BNET_ENHANCED_MODE_ON, MDS_ENHANCED_MODE_ON, PLUS_ISO_ENHANCED_MODE_ON and VISA_ENHANCED_MODE_ON must be set to TRUE.

For a release 6.0 BASE24 network environment supporting Switch 6.0 Banknet, EPS-Net, MDS/Cirrus ISO, MDS Maestro Gateway or VisaNet ISO Interfaces, the respective CUSTMACS flags BNET_EMV_ON, EURO_EMV_ON, MDS_ISO_EMV_ON, MDS_MAESTRO_EMV_ON and VISA_EMV_ON must be set to TRUE

if you want the interface to support EMV processing. Otherwise, the flags must be set to FALSE.

LCONF Parameter

SECURE-DEV-TYP

A code identifying the type of security module specified in the SECURE-DEV1 and SECURE-DEV2 assigns. Valid values are:

- ASM = Atalla Security Module or Atalla Network Security Processor
- RSM = Racal-Guardata Host Security Module (same as VSM)
- VSM = Racal-Guardata Host Security Module (same as RSM)
- TSS = Transaction Security Services
- N/A = A security module is not required (default)

This parameter must have the value 'TSS' if dynamic key management and/or triple DES encryption (3DEA) are to be supported via Transaction Security Services.

DES-TRIPLE-SINGLE

This param will let the interface know whether triple to single DES encryption processing is required or not. Valid values include the following:

Y = Triple to single DES encryption processing is required.

N = Triple to single DES encryption processing is not required.

This parameter must be set to 'N' if it exists in order for the interface to utilize Transaction Security Services. This parameter must exist and be set to 'Y' if triple to single DES encryption processing is to be supported without Transaction Security Services.

LCONF Assigns

APCFEMT

The physical file name of the Acquirer Processing Code File Extended Memory Table. If you are running a Switch 6.0 Interface that has been enhanced to support the Transactions Allowed Enhancement (Banknet, BIC ISO, MDS/Cirrus ISO, MDS Maestro Gateway, PLUS ISO or VisaNet ISO), the ICFE assign must be present in the LCONF. This assign is required when running the above mentioned Switch 6.0 Interfaces in a BASE24 release 6.0 network environment.

ICFE

The physical file name of the Enhanced Interchange Configuration File. If you are running a Switch 6.0 Interface that has been enhanced to support the Transactions Allowed Enhancement (Banknet, BIC ISO, MDS/Cirrus ISO, MDS Maestro Gateway, PLUS ISO or VisaNet ISO), the ICFE assign must be present in the LCONF. This assign is required when running the above mentioned Switch 6.0 Interfaces in a BASE24 release 6.0 network environment.

IPCFEMT

The physical file name of the Issuer Processing Code File Extended Memory Table. If you are running a Switch 6.0 Interface that has been enhanced to support the Transactions Allowed Enhancement (Banknet, BIC ISO, MDS/Cirrus ISO, MDS Maestro Gateway, PLUS ISO or VisaNet ISO), the ICFE assign must be present in the LCONF. This assign is required when running the above mentioned Switch 6.0 Interfaces in a BASE24 release 6.0 network environment.

SECURE-DEV1

The Tandem SYSGEN device name of the primary security module. However, if the interface is running in a release 6.0 network environment using Transaction Security Services, this parameter must contain the PPD name of the primary IAS process and may be process specific. This assign is required.

SECURE-DEV2

The Tandem SYSGEN device name of the secondary security module. However, if the interface is running in a release 6.0 network environment using Transaction Security Services, this parameter must contain the PPD name of the secondary IAS process and may be process specific. This assign is optional.

System Impacts

The following impacts to the system are caused by implementing Switch 6.0 Interfaces.

Disk Space Impacts

Additional disk space may be required to support Switch 6.0 Interfaces. Refer to the interface update and manual files on the respective interface subvolume for a detailed description on disk space impacts. The introduction of new tokens that may be logged to the ILF, TLF or PTLF files may impact disk space requirements. Additional functionality that increases or decreases the size of a file or requires new files may also impact disk space requirements.

RELEASE 6.0 SOURCE CODE IMPACTS

Switch 6.0 Interface source code is cataloged and sent out on release 6.0 subvolumes (\$vol.SW60xxxx) for each Switch 6.0 Interface even if the customer is running a release 3.4/4.0, 5.0, 5.1.2, or 5.3 BASE24 network.

ACI also sends customers the release 6.0 base source code which resides on the BA60AFT, BA60CSE, BA60DDL, BA60DDOC, BA60DEFS, BA60EMTB, BA60IS92, BA60LIB, BA60LOGM, BA60MAKE, BA60MISC, BA60SRC, BA60UB04, BA60UC04, BA60UD04 and BA10SC04. All Switch 6.0 Interfaces use the release 6.0 base source code so that interface modules, interchange reports, and release 6.0 Pathway modules make correctly. Customers currently on BASE24 release 5.3 or earlier must maintain base subvolumes on two different volumes, one for release 6.0 base code and one for their current version of base code.

ACI also sends customers the release 6.0 kernel code which resides on the OK60AFM, OK60DDL, OK60ISMT, OK60LIB, OK60ONCM, OK60PCTL and OK60SRC subvolumes. Some of the Switch 6.0 Interfaces use the release 6.0 kernel source code so that interface modules make correctly.

DDL Impacts

ACI will send the release 6.0 BA60DDL and OK60DDL subvolumes to its customers along with Switch 6.0 Interface code. Releases 3.4/4.0, 5.0, 5.1.2 and 5.3 BASE24 customers implementing Switch 6.0 Interfaces will make the Switch 6.0 Interface in the release 6.0 MAKE stream, utilizing the release 6.0 base and kernel code.

To improve the readability and usability of BASE24 files and record structures, ACI provides a utility program called DDLDOC. DDLDOC retrieves information from the compiled data dictionary and formats the information as the user requests. The information provided by DDLDOC includes the field's position in the file, the field name, and the data type of the field. Field names can be shown in COBOL or TAL format. For more information on the DDLDOC utility, refer to the "BASE24 Files and Record Structures Reference Manual".

The ICFE (Enhanced Interchange Configuration File), APCF (Acquirer Processing Code File) and IPCF (Issuer Processing Code File) are new files that will be utilized by the Switch 6.0 Interfaces that have been enhanced to support the Transactions Allowed Enhancement (BIC ISO, Banknet, MDS/Cirrus ISO, MDS Maestro Gateway, PLUS ISO and VisaNet ISO). The Transactions Allowed Enhancement is only supported in a release 6.0 BASE24 network environment. Therefore, the ICFE will only be utilized by the above mentioned Switch 6.0 Interfaces in a release 6.0 BASE24 network environment. If the interfaces are being run in a pre-release 6.0 BASE24 network environment, they will utilize the ICF (Interchange Configuration File).

The KEY6 file is a new file that may be utilized by the Switch 6.0 Interfaces that have been modified to support the Triple/Single DES Enhancement, (BIC ISO, Banknet, MDS/Cirrus ISO, MDS Maestro Gateway and VisaNet ISO). If these Switch 6.0 Interfaces are configured to support triple to single DES encryption without Transaction Security Services, they will utilize the KEY6. Otherwise, they will utilize the KEYF (Key File). Triple to single DES encryption is supported within these interfaces using the KEY6 in any BASE24 release network environment. Triple DES encryption is supported via Transaction Security Services within these interfaces using the KEYF only in a release 6.0 network environment.

The ISEC file is a new file that may be utilized by the Switch 6.0 Interfaces that have been modified to support triple DES encryption (3DEA) via Transaction Security Services, (Banknet, BIC ISO, MDS/Cirrus ISO, MDS Maestro Gateway and VisaNet ISO). Triple DES encryption via Transaction Security Services is supported within these interfaces only in a BASE24 release 6.0 network environment.

Interchange Interface Report Impacts

One set of report programs is used by all Switch 6.0 Interfaces unless interchange specific information is required. This set of report programs is located on the Switch 6.0 Interface reports subvolume (SW60RPT) which is sent to customers by ACI with the release 6.0 code. The report programs on SW60RPT generate the Interchange Clearings Statement Report (SETL01) and the Interchange Transaction Detail Report (STAT09). All of the source code on this subvolume is required so that the release 6.0 generic interchange interface reports will compile correctly.

If the generic report programs require interchange-specific changes, these interchange specific report programs are placed on the interface subvolume.

Interchange Log File (ILF) Impacts

RELEASE 4.0 IMPACTS

EXTRACT

Customers currently running Fortify interfaces in a release 3.4/4.0 BASE24 network, can continue to use their 4.0 version of the ILF extract for Switch 6.0 Interfaces. When ACI created the original Fortify, a fix to the 3.4/4.0 ILF Extract to source in the 5.0 ILF DDL was provided to customers. This fix still applies and customers can continue to run the 4.0 ILF Extract.

If Switch 6.0 Interfaces are running in a release 3.4/4.0 BASE24 network, the release 3.4/4.0 ILF extract module sources in the release 5.0 ILF DDL. This change was provided to customers as a fix along with the Fortify code. This change to the release 3.4/4.0 ILF extract source file, named ILFXS, and the BASE24 release 5.0 source code subvolume, named BA50SRC, must reside on the same Tandem volume so that the ILF extract file sources in the correct ILF DDL (i.e., ILF and ILFX from BA50SRC.BADDLTAL) for compilation.

RELEASE 5.0 IMPACTS

Customers running Switch 6.0 Interchange Interfaces in a BASE24 release 5.0 system should continue to use the release 5.0 ILF DDLs for ILF Extract since no structural changes have been made to the ILF DDLs.

Host Program Impacts

Customers currently supporting Fortify.1 or Fortify.3 should not have to make host changes due to the Switch 6.0 release unless noted in the SyncPt 04 Implementation Guide. Customers should evaluate support for tokens or new tokens as they move forward with 5.0 and greater releases of BASE24.

RISC Impacts

The MAKE files provided to customers with the Switch 6.0 Interfaces contain statements to initiate the Reduced Instruction-Set Computing (RISC) Accelerator. If RISC instructions are desired, the accelerator flag SWxxxx_AXCEL_ON may be set to TRUE in the configuration MAKE file (xxxxFM) located on the interface subvolume, where "xxxx" identifies the interface. For more information on RISC, refer to the "RISC Implementation Guide".

ICF/ICFE REQUESTER/SERVER PAIR IMPACTS

As customers integrate Switch 6.0 Interfaces into their system, they will be provided with release 6.0 interchange-specific requester/server pairs located on the interchange-specific subvolume with the code. These interchange-specific requester/server pairs allow customers to access the interchange-specific ICF/ICFE screens for the particular interface.

EMF/KEYF Requester/Server Pair Impacts

Customers running Switch 6.0 Interfaces in a 4.0 network must set the FORTIFY_ON flag to TRUE in the CUSTMACS before running MAKE. This will cause MAKE to create a 4.0 version and a 5.0 version of the EMF and KEYF requesters. The EMF 4.0 requester will be named T6520-EMF and T3270-EMF and the EMF 5.0 requester will be named T6520-EMF5 and T3270-EMF5. The KEYF 4.0 requester will be named T6520-KEYF and T3270-KEY5 and the KEYF 5.0 requester will be named T6520-KEY5 and T3270-KEY5.

KEY6 Requester/Server Pair Impacts

As customers integrate Switch 6.0 Interfaces into their system, they will be provided with a release 6.0 KEY6 requester/server pair located on the BA60AFT subvolume. This requester/server pair is utilized to access the KEY6 file that contains the double length keys needed to encrypt PINs and keys for triple to single DES encryption without Transaction Security Services.

File Conversion Programs

Conversions for Switch 6.0 Interfaces

Determine what mandates/enhancements you are currently running. You must run all conversion programs on the interface subvolume with a date greater than the latest mandate/enhancement in your production interface.

ACI supplies the ILF and IIF conversion programs for those switches affected. The procedures for running conversion programs are included on each switch subvolume in an update guide or manual file.

NO CONVERSIONS ARE NEEDED IF THE CUSTOMER HAS ALREADY IMPLEMENTED THE ENHANCEMENTS OR IF IT IS A NEW INSTALLATION OF A SWITCH 6.0 INTERFACE.

Enhanced Interchange Configuration File (ICFE) Conversion

The Banknet, BIC ISO, MDS/Cirrus ISO, MDS Maestro Gateway, PLUS ISO and VisaNet ISO Interfaces require an ICFE file conversion that will convert the release 5.3 ICF records to release 6.0 ICFE records. The above mentioned interfaces utilize the ICFE instead of the ICF in a release 6.0 network environment in order to support the Transactions Allowed Enhancement.

ACI supplies the ICFE conversion program for those switches affected. Create a new ICFE file by using the BA60SRC.BADDLFUP file. Refer to the run file BA60UC04.ICFER for the ICFE conversion program execution details.

The ICFE file conversion is necessary due to DDL changes required for enhancements to the above mentioned Switch 6.0 Interfaces. NO CONVERSION IS NEEDED IF NONE OF THE ABOVE MENTIONED SWITCH 6.0 INTERFACES ARE BEING IMPLEMENTED.

KEYF/KEY6 CONVERSIONS

In order to support dynamic key management via Transaction Security Services, the Alaska Option ISO, ANSI BASE24, Cash Station ISO, Deluxe ISO Generic, MAC MASM, Money Payment Systems Host to Host, Money Station, NEN/Excel ISO, NPC ISO, Plus ISO, Publix and Shazam ISO Interfaces require a KEYF file conversion that will convert the current KEYF records to the release 6.0 KEYF records.

In order to support dynamic key management and triple DES encryption via Transaction Security Services, the Banknet, BIC ISO, MDS/Cirrus ISO, MDS Maestro Gateway, and VisaNet ISO Interfaces require either a KEYF or KEY6 file conversion that will convert the current KEYF or KEY6 records to the release 6.0 KEYF records. If the customer is currently supporting any of the above mentioned interfaces with the KEYF, a KEYF conversion is required. If the customer is currently supporting any of the above mentioned interfaces with the KEY6, a KEY6 conversion is required.

The KEYF file conversion is necessary due to manner in which KEYF data is utilized in the above mentioned Switch 6.0 Interfaces. The KEY6 file conversion is necessary because triple DES encryption via Transaction Security Services utilizes the KEYF instead of the KEY6. The conversion(s) is required if Transaction Security Services is configured in the BASE24 release 6.0 network environment. The entire system must be configured to utilize Transaction Security Services if any module requires Transaction Security Services.

ACI supplies the KEYF conversion program for those switches affected. Create a new KEYF file by using the BA60SRC.BADDLFUP file. Refer to the run file BA60UC05.TSSKEYFR for the KEYF conversion program execution details.

ACI supplies the KEY6 conversion program for those switches affected. Create a new KEYF file by using the BA60SRC.BADDLFUP file, if you have not already done so. Refer to the run file BA60UC05.TSSKEY6R for the KEY6 conversion program execution details.

File Impacts

Several new files are required to support Switch 6.0 Interfaces. For more information on the DDLs used with Switch 6.0 Interfaces, refer to the "DDL Impacts" topic in the System Impacts section in this document.

BASE SUBVOLUMES

Subvolume BA60EMTB was added in order to support EMT Build. Subvolume BA60LIB was added in order to support release 6.0 library routines. Files from both of these subvolumes are utilized by the Switch 6.0 Interfaces.

Interchange Log File (ILF) Impacts

ACI supplies conversion programs to convert previous ILF records to new ILF records and ACI recommends using these programs to ensure all ILF records are properly copied to the release 6.0 ILF. For information on the ILF conversion programs, refer to the "File Conversions" topic in this section and the SW60xxxx.xxxxUPDT where "xxxx" identifies the interface.

Format 2 Files support, will be available in the Switch 6.0 Banknet, BIC ISO, EPS-Net, LINK, MDS/Cirrus ISO, MDS Maestro Gateway, Star ISO, VisaNet ISO, VisaNet Interlink Gateway and VisaNet/PLUS Gateway Interfaces. Format 2 Files support requires a release 6.0 BASE24 environment and a D46 or higher Compaq NSK operating system. Format 2 Files support allows customers the ability to format Enscribe disk file (ILF) partitions to accommodate Format 2 files. For information on implementing Format 2 File ILFs in the above mentioned Switch 6.0 Interfaces, refer to the specific SW60xxxx.xxxxUPDT file where "xxxx" identifies the interface.

Institution Identification File(IIF) Impacts

ACI supplies a conversion program to convert previous IIF records to new IIF records for the VisaNet ISO Interface. ACI recommends using this program to ensure all IIF records are properly copied to the release 6.0 IIF. For information on the IIF conversion program, refer to the "File Conversions" topic in this section and the VISAUPDT file.

3: Reference Material

Interchange Reports

The Switch 6.0 Interchange reports are used with Switch 6.0 Interchange Interfaces. All BASE24 release customers receive the Switch 6.0 Interchange report subvolume (SW60RPT) in its entirety so Switch 6.0 Interchange reports can compile correctly. No modifications are required to the Switch 6.0 Interchange report programs to allow them to run in a release 3.4/4.0/5.0/5.1.2/5.3/6.0 BASE24 network.

Log Message Documentation

For release 6.0, log message documentation on the Tandem resides on the switch subvolumes. Log message files are named xxxxLOGM where "xxxx" identifies the interface. Each log message file begins with a history section. The documentation includes the log message and its assigned log message number, definitions of all variables contained in the log message, the cause of the log message, and the recommended action to take when the log message is encountered.

Manuals and Update Files

Each switch has a manual describing installation, parameters, files, commands and reporting. The manual can be found under the name xxxxMAN where "xxxx" identifies the interface. Some information from old update files has been rolled into these manuals.

Update guides have been provided for Interchange Interfaces that have been significantly changed due to enhancements. The guides can be found under the name xxxxUPDT where "xxxx" identifies the interface. The xxxxUPDT guide provides detailed information about each enhancement and its impacts.